

Advance Bash Scripting-2

String Manipulation

1. Basic Declaration:

```
#!/bin/bash
text="Welcome to PW SKILLS"
echo "Message: $text"
```

2. Extract String

syntax: \${string:position:length}

```
#!/bin/bash
text="welcome to PW SKILLS"
echo "OG String: $text"
echo "Substring (start at 15, length 6): ${text:14:6}"
```

3. Replace SubString

- Replace First occurrence: \${string/old/new}
- Replace all occurrences: \${string/old/new}

```
#!/bin/bash
message="hello world world"
echo "Replace first: ${message/world/Bash}" # hello Bash World
echo "Replace all: ${message//world/Bash}" # hello Bash Bash
```

4. Find Length of String

```
#!/bin/bash
mystr="My Name is Nikunj And I am DevOps Trainer"
echo "Length: ${#mystr}" # Output:41
```

5. Extract from End (Negative Index)

- Bash doesn't directly supports negative index but we can simulate it:

```
#!/bin/bash
str="ABCDEFGH"
str1="Hello World"
echo "Last 3 Characters:${str: -3}" # Note the Space After:
echo "Last 5 Characters: ${str: -5}"
```

6. Convert Case

- using string manipulation we can convert a string to Uppercase,lowercase

```
#!/bin/bash
name="nikunj soni"
echo "Uppercase: ${name^^}" # NIKUNJ
echo "Capitalize: ${name^}" # Nikunj
str="TRAINER"
echo "lowercase: ${str,,}" #trainer
```

1. Remove Prefix or Suffix

- Remove shortest match:
 - Prefix: \${string#pattern}
 - Suffix: \${string%pattern}
- Remove longest match
 - Prefix: \${string##pattern}
 - Suffix:\${string%%pattern}

```
#!/bin/bash
file="photo.jpg"

echo "Remove Shortest Suffix: ${file%.*}"
echo "Remove extension: ${file%%.*}"
```

8. Concatenate Strings

- used to join the strings

```
#!/bin/bash
first="Good"
second="Night"
greeting="$first $second"
echo "$greeting" #output: Good Night
```

9. Check if Substring Exist or not

```
#!/bin/bash
text="Welcome to Linux World"
if[ [ "$text" == *"Linux"* ] ]; then
echo "Found Linux"
```

fi



Task: Parse and Format User Information



Problem Statement

You are given a string containing a user's details in the following format:

```
perl
CopyEdit
"full_name=nikunj soni;email=nikunj.soni@example.com;city=ahmedabad"
```

Write a Bash script that:

1. Extracts each field (full_name, email, city).
2. Converts the full name to **uppercase**.
3. Extracts the **username** from the email (i.e., part before @).
4. Capitalizes the first letter of the city.
5. Outputs the formatted data as:

Solution:

```
#!/bin/bash
#String
input="full_name=nikunj soni;email=nikunj.soni@example.com;city=ahmedabad"

# Split the string by ';'
IFS=';' read -ra fields <<< "$input"

# Initialize variables
full_name=""
email=""
city=""

# Extract values
for field in "${fields[@]}"; do
    key="${field%%=*}"
    value="${field#*=}"
    case $key in
        full_name) full_name="${value^^}" ;; # Uppercase full name
        email) email="$value" ;;
        city) city="${value^}" ;; # Capitalize city
    esac
done
```

```
esac  
done
```

```
# Extract username from email  
username="${email%@*}"
```